

Real-World Examples of Inheritance

Let's take a look at two real-world examples of inheritance. First, I'll show you how to build an input field that only allows you to enter numbers. I'll then build an output field that formats monetary values.

InputNumber

In this first example, we'll create an input field that allows the user to input only numerical data. As a general user interface design principle, it's much better to restrict the input to allow only for correctly formatted data while the user is entering the data, rather than allowing any input and checking its correctness later. Entering letters or other symbols in this input field, or attempting to enter multiple decimal points or multiple minus signs, should not be allowed.

The `pygwidgets` package contains an `InputText` class that allows the user to input any characters. We'll write an `InputNumber` class to allow only valid numbers as input. The new `InputNumber` class will inherit much of its code from `InputText`. We'll only need to override three methods of `InputText`: `__init__()`, `handleEvent()`, and `getValue()`. [Listing 10-3](#) shows the `InputNumber` class that overrides these methods.

File: MoneyExamples/InputNumber.py

```
# InputNumber class - allows the user to enter only numbers
#
# Demo of inheritance

import pygame
from pygame.locals import *
import pygwidgets

BLACK = (0, 0, 0)
WHITE = (255, 255, 255)
# Tuple of legal editing keys
LEGAL_KEYS_TUPLE = (pygame.K_RIGHT, pygame.K_LEFT, pygame.K_HOME,
                    pygame.K_END, pygame.K_DELETE, pygame.K_BACKSPACE,
                    pygame.K_RETURN, pygame.K_KP_ENTER)
# Legal keys to be typed
```